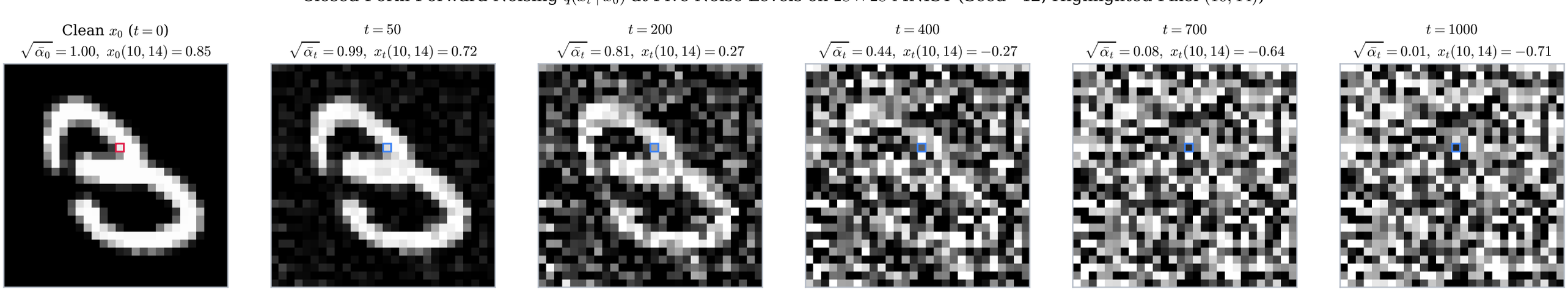
Closed-Form Forward Noising $q(x_t \mid x_0)$ at Five Noise Levels on 28×28 MNIST (Seed=42, Highlighted Pixel $(10, 14)$)
Clean x_0 $(t = 0)$
$\sqrt{\bar{\alpha}_0} = 1.00,\ x_0(10, 14) = 0.85$
$t = 50$
$\sqrt{\bar{\alpha}_t} = 0.99,\ x_t(10, 14) = 0.72$
$t = 200$
$\sqrt{\bar{\alpha}_t} = 0.81,\ x_t(10, 14) = 0.27$
$t = 400$
$\sqrt{\bar{\alpha}_t} = 0.44,\ x_t(10, 14) = -0.27$
$t = 700$
$\sqrt{\bar{\alpha}_t} = 0.08,\ x_t(10, 14) = -0.64$
$t = 1000$
$\sqrt{\bar{\alpha}_t} = 0.01,\ x_t(10, 14) = -0.71$